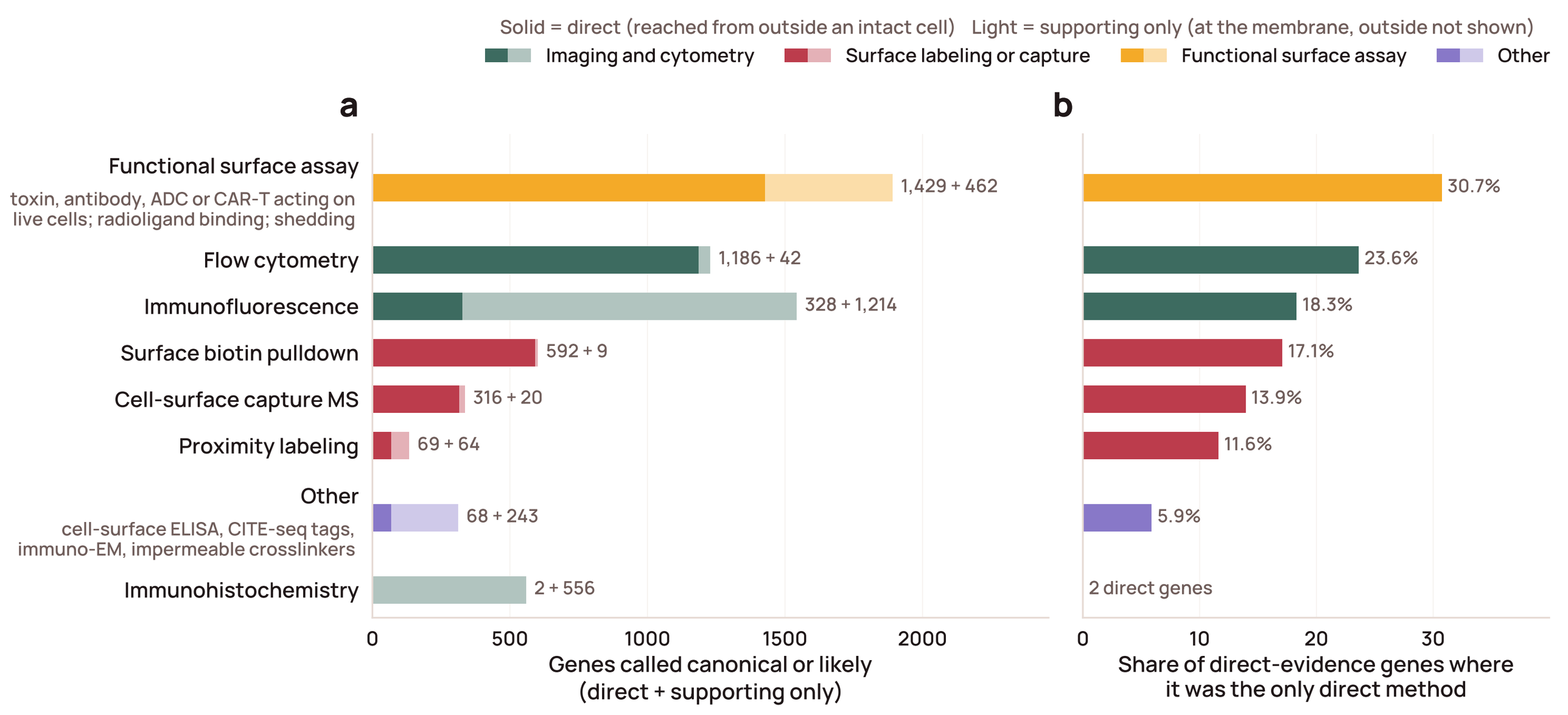
Solid = direct (reached from outside an intact cell)
Light = supporting only (at the membrane, outside not shown)
Imaging and cytometry
Surface labeling or capture
Functional surface assay
Other
a
b
Functional surface assay
toxin, antibody, ADC or CAR-T acting on live cells; radioligand binding; shedding
Flow cytometry
Immunofluorescence
Surface biotin pulldown
Cell-surface capture MS
Proximity labeling
Other
cell-surface ELISA, CITE-seq tags, immuno-EM, impermeable crosslinkers
Immunohistochemistry
1,429 + 462
1,186 + 42
328 + 1,214
592 + 9
316 + 20
69 + 64
68 + 243
2 + 556
0
500
1000
1500
2000
Genes called canonical or likely (direct + supporting only)
30.7%
23.6%
18.3%
17.1%
13.9%
11.6%
5.9%
2 direct genes
0
10
20
30
Share of direct-evidence genes where it was the only direct method